Grade 9 Biology Annual plan

Name of School: Jigjiga Special Boarding School
Teacher Name: Girmaw Asaye
Subject Biology
Grade 9
Academic year: 2018 E.C

Total Number of learning days: 214
Total number of learning days of first semester: 112
Total number of learning days of second semester: 102
Total periods of first semester: 63
Total periods of second semester: 57

Semester	Months	Weeks	Dates	Contents/Main Topics and subtopics	General objectives At the end of this unit, learners will be able to:	Methodology	Instructional media	Assessment	
								Formative	Summative
First Semester	September	1	05 – 09	Unit:1 Introduction to Biology 1.1 Definition of Biology	Explain the nature and importance of biology Define Biology	Inquiry based teaching through Discussion Demonstration Lecture Mind mapping Debates	Hand lenses Microscope test tubes autoclave incubators, petri dish flasks, balance	Reflection, observation of students' during discussion Presentation,	Homework Class work Written test Quiz
		2	12 – 16	1.2 Why do we study Biology	Explain why Biology is studied				
		3	19 – 23	1.3 The scientific method	Plan a biological investigation using the scientific method				
		4	26 – 30	1.4 Tools of a Biologist 1.4.1 Laboratory tools 1.4.2 field tools	Identify some common Laboratory and field tools of a biologist Explain the function of biological tools				
	October	1	03 – 07	1.5 The light Microscope 1.5.1Parts and functions of light microscope 1.5.2 handling and using light microscope 1.6 General Laboratory Safety Rules	Define light Microscope Identifying parts of microscope Utilize a microscope Execute general laboratory safety rules				
		2	10 – 14	Unit: 2 Characteristics and classification of organisms 2.1 Characteristics of living things 2.2 Taxonomy of living things	Explain what is life State the characteristics of living things Classify living things based on taxonomic principles Describe taxonomic hierarchies	Inquiry based teaching through Discussion Demonstration Lecture Mind mapping Debates	Real models of different plants and animals Pictures and diagrams	Reflection, observation of students' during discussion Presentation	Homework Class work Written test Quiz
		3	17 – 21	2.3 Relevance of classification 2.4 Linnaean system of nomenclature	Argue for or against the importance of classification Describe the system of Linnaean nomenclature				
		4	24 – 28	2.5 Common Ethiopian animals and plants 2.6 The five kingdom system of classification	Classify common Ethiopian animals and plants based on the taxonomic categories. List the characteristics features of the five kingdom				
	November	1	01 – 05	2.6.1 Kingdom Monera 2.6.2 Kingdom Protista 2.6.3 Kingdom fungi	Describe the features of kingdom Monera, Protista and fungi with in examples				
		2	08 – 12	2.6.4 Kingdom Plantae 2.6.5 Kingdom animalia 2.7 Renowned Taxonomist in Ethiopia	Describe the features of kingdom animalia Appreciate the works of renowned taxonomists in Ethiopia				
		3	15 – 19	Unit : 3 Cells 3.1 What is cell 3.2 Cell theory	Define a cell Outline the contributions of Theodor Schwann and Matthias Schleiden to the development of the cell theory	Inquiry based teaching through Discussion Demonstration Lecture Mind mapping Debates	Diagrams of cells Microscope Prepare slide of plant and animal cells	Observe students discussion and Presentation	Homework Class work Written test Quiz
		4	22 – 26	3.3 Cell structure and function	Compare and contrast the structures of plant and animal cells				

Second Semester	December	1	29 – 03	3.4 Types of cells	Differentiate b/n eukaryotic and prokaryotic cells	Inquiry based teaching through Discussion Demonstration Lecture Mind mapping Debates	Potato Onion Salt and water	Reflection, observation of students' during discussion Presentation	Homework Class work Written test Quiz
		2	06 – 10	3.5 animal and plant cells 3.6 Observing cells under microscope	Make temporary slides of plant and animal cells Observe and draw animal and plant cell parts under the microscope				
		3	13 – 17	3.6 Observing cells under microscope 3.7 The cell and its environment	Observe and draw animal and plant cell parts under the microscope				
		4	20 – 24	3.7 The cell and its environment 3.7.1 Passive transport 3.7.2 Active transport	Conduct substances across animal and plant cells experiments about the movement of substances across animal and plant cells (osmosis, diffusion)				
		5	27 – 01	3.8 Level of biological organizations Unit: 4 Reproduction 4.1 Introduction to reproduction 4.2 Asexual Reproduction	Construct with examples the different levels of biological organization Compare and contrast the advantages and dis advantages of asexual and sexual reproduction	Inquiry based teaching through: Discussion Demonstration Lecture Mind mapping Debates	Diagram of male and female reproductive organ	Reflection, observation of students' during discussion Presentation	Homework Class work Written test Quiz
	January	1	04 – 08	4.3 Types of Asexual reproduction 4.3.1 Fission 4.3.2 fragmentation 4.3.3 Budding	Define asexual and sexual reproductions Describe asexual reproductions in bacteria, fungi, animals, and plants				
		2	11 – 15	4..3.4 vegetative propagation 4.3.5 Parthenogenesis	Investigate the mechanism of vegetative propagation Describe the mechanism of Parthenogenesis				
		3	18 – 22	First Semester Final Exam					
		4	25 – 29	Roster Preparation					
Second Semester	February	1	02 – 06	4.4 Sexual reproduction in humans 4.5 primary & Secondary sexual characteristics	Differentiate between the primary and secondary sexual characteristics of males and females in humans	Inquiry based teaching through: Discussion Demonstration Lecture Mind mapping Debates	Diagram of male and female reproductive organ Real materials of Birth controlling devices	Reflection, observation of students' during discussion and presentation	Homework Class work Written test Quiz
		2	09 – 13	4.6 Male reproductive structure 4.7 female reproductive structure	Identify the structure and functions of the Male reproductive organs. Identify female reproductive structures and their functions				
		3	16 – 20	4.7 Female reproductive structure 4.8 Menstrual cycle	Identify female reproductive structures and their functions Outline the phases of the menstrual cycle				
		4	23 – 27	4.9 Fertilization and pregnancy 4.10 Methods of birth control (Natural method of family planning)	Define fertilization and pregnancy Discuss the methods of prevention of pregnancy				

Second Semester	March	1	30 – 04	4.10 Method of birth control (Artificial methods of family planning)	Discuss the methods of prevention of pregnancy	Discussion Demonstration Mind mapping Debates	Real materials of Birth controlling devices	Give feedback	Classwork Homework
		2	07 – 11	4.11 Sexual transmitted infections (sits)	Describe the types, modes of transmission and preventive mechanisms of sexually transmitted diseases				
		3	14 – 18	Unit: 5 Human health, nutrition and disease 5.1 What is food 5.2 Nutrition 5.3 Nutrients (Carbohydrate)	Define nutrition List the types of nutrients	Inquiry based teaching through: Discussion Demonstration Lecture Mind mapping Debates	Different food samples, cigarettes, Chat Diagrams of HIV/AIDS, Covid 19 etc	Reflection, observation of students' during discussion Presentation	Homework Class work Written test Quiz
		4	21 – 25	5.3 Nutrients (Lipids and proteins, Micronutrients, water and fiber)	Describe the role of a balanced diet especially for children, nursing mothers, athletes, and people with HIV/AIDS				
		5	28 – 02	5.4 Balanced diet 5.5 Deficiency disease	Discuss the feeding habits that may lead to obesity				
Second Semester	April	1	05 – 09	5.5 Deficiency disease (Causes) 5.6 Malnutrition	Describe with examples the sources and deficiency diseases of vitamins and minerals. Define malnutrition				
		2	12 – 16	5.7 Substance abuse - Substance abuse, cigarettes smocking - Alcohol drink & Heroin - Cannabis & Chewing Chat	Examine the effects of smoking, alcohol use, chewing chat, cannabis, and other drug use, on the health, social, economic, cultural, and psychological wellbeing				
		3	19 – 23	5.5 Substance abuse - Doping	Examine the effects of Doping on health				
		4	26 – 30	5.8 Infectious and noninfectious diseases 5.8.1 Infectious disease - HIV/AIDS and TB - Malaria - Covid – 19	Explain the modes of transmission and prevention of infectious and noninfectious diseases				
	May	1	03 – 07	5.8.2 Noninfectious disease 5.9 Renowned Nutritionists in Ethiopia	List and appreciate renowned nutritionists in Ethiopia				
		2	10 – 14	Unit: 6 Ecology 6.1 Definition of ecology 6.1.1Deffnation of common ecological terms 6.1.2 Biotic and Abiotic components	Define ecological terms Explain the biotic and abiotic components of an ecosystem Express love and respect to fauna and flora and their biomes	Inquiry based teaching through: Discussion Demonstration Lecture Mind mapping Debates	Living organisms such as plants, animals Abiotic factors such as water and soil Different landscape diagrams	Reflection, observation of students' during discussion Presentation	Homework Class work Written test Quiz
		3	17 – 21	6.1.3 Ecological levels 6.1.4 Ecosystem	Describe the ecological levels Explain the terrestrial and aquatic ecosystem				
		4	24 – 28	6.1.5 Biomes - Exploring major terrestrial biomes	Define biome Describe the major terrestrial and aquatic biomes				
	June	1	01 – 05	6.1.5 Biomes - Exploring major aquatic biomes	Mention the fauna and flora of each biome Describe the major major aquatic biomes				
		2	08 – 12	6.1.6 Ecological succession 6.2 Ecological relationships	Describe ecological succession Discuss ecological relationships in Ecosystem				
		3	15 – 19	Second Semester Final Exam					
		4	22 – 26	Roster Preparation					

Teacher's Name _____ Date _____ Director's Name _____ Date _____

Signature _____ Signature _____